CS229
Problem Set 2

Connor Yan

April 27, 2026

Problem 1

(a) Regression on dataset B does not converge.

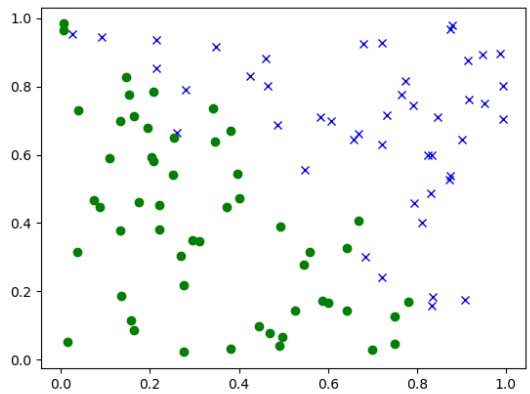

Figure 1: Dataset A

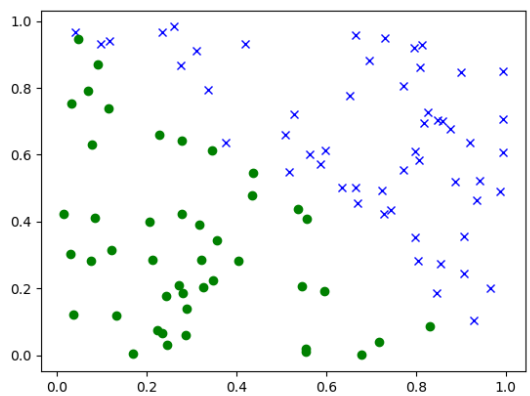

Figure 2: Dataset B

(b) Notice that dataset B is linearly separable, whereas dataset A is not. The loss function of this logistic regression is

$$\ell(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\log\left(\frac{1}{1+e^{-y^{(i)}\theta^T x^{(i)}}}\right) = \frac{1}{m}\sum_{i=1}^{m}\log\left(1+e^{-y^{(i)}\theta^T x^{(i)}}\right)$$

As a negative log-likelihood, this is definitely greater than or equal to zero. However, for linear separable cases, if the fetched θ is good enough to rightly separate all data points, then this θ can be scaled up by an arbitrary positive number, and the classification is still right, with a smaller loss function. As $||\theta|| \to \infty$,

$$\ell(\theta) \to \frac{1}{m}\sum_{i=1}^{m}\log\left(1+e^{-\infty}\right) \to 0,$$

which means there is not a finite θ that yields to the minimum of $\ell(\theta)$.

Problem 1 continued on next page. . .

For non-linear-separable cases, where $y^{(i)}\theta^T x^{(i)} < 0$, terms in the loss function corresponding to data points that cannot be rightly classified are

$$\log(1 + e^{-y^{(i)}\theta^T x^{(i)}}),$$

As $||\theta|| \to \infty$,

$$\log(1 + e^{-y^{(i)}\theta^T x^{(i)}}) \to \infty$$

This indicates there exists a θ that gives the minimum of the loss function, namely, the gradient descent can converge.

(c)
i. No. This is just a part of the gradient descent and does nothing with whether the minimum of the loss function exists.
ii. No. This is just a part of the gradient descent and does nothing with whether the minimum of the loss function exists.
iii. No. This change will not affect whether the dataset is linearly separable or not.
iv. Yes. This can prevent infinitely scaling θ up to decrease the loss function.
v. Probably would work if the modified dataset becomes non-linear-separable.

(d) No. SVMs minimize $\frac{1}{2}||w||^2$, which punishes inifinitely scaling up of w.

Problem 2

(a) Logistic regression is to maximise the log-likelihood

$$\ell(\theta) = \sum_{i=1}^{m} y^{(i)} \log(h_\theta(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_\theta(x^{(i)}))$$

Therefore, after training, it is supposed that

$$\begin{aligned}\nabla\ell(\theta) &= \sum_{i=1}^{m} [y^{(i)}(1 - h_\theta(x^{(i)})) + (1 - y^{(i)})h_\theta(x^{(i)})]x^{(i)} \\ &= \sum_{i=1}^{m} [y^{(i)} - h_\theta(x^{(i)})]x^{(i)} = 0\end{aligned}$$

Thus,

$$\sum_{i=1}^{m} y^{(i)} - \sum_{i=1}^{m} h_\theta(x^{(i)}) = 0$$

$$\frac{\sum_{i=1}^{m} h_\theta(x^{(i)})}{m} = \frac{\sum_{i=1}^{m} y^{(i)}}{m}$$

Namely,

$$\frac{\sum_{i \in I_{0,1}} P\left(y^{(i)} = 1 \mid x^{(i)}; \theta\right)}{|\{i \in I_{0,1}\}|} = \frac{\sum_{i \in I_{0,1}} \mathbf{1}\left\{y^{(i)} = 1\right\}}{|\{i \in I_{0,1}\}|}$$

(b) Both are not true.

Consider a set of examples where 70% of the examples are positive, and the hypothesis is 0.7 for all examples. It will perfectly satisfy the property, but eventually will predict that all examples are positive.

For the converse, consider a classifier that can perfectly classify a dataset with 70% positive cases, and its hypothesis function equals 0.9 for all positive examples and equals 0.1 for all negative examples. Then if we let $(a, b) = (0.5, 1)$, the average hypothesis is 0.9, whereas the frequency of positive examples is 1.

Problem 2 continued on next page. . .

(c) The log-likelihood with L_2 Regularization is

$$\ell(\theta) = \left[\sum_{i=1}^{m} y^{(i)} \log(h_\theta(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_\theta(x^{(i)}))\right] - \lambda\|\theta\|^2$$

An ideally trained model should have the gradient

$$\nabla\ell(\theta) = \left[\sum_{i=1}^{m} [y^{(i)} - h_\theta(x^{(i)})]x^{(i)}\right] - 2\lambda\theta = 0.$$

Hence,

$$\frac{\sum_{i=1}^{m} h_\theta(x^{(i)})}{m} \neq \frac{\sum_{i=1}^{m} y^{(i)}}{m},$$

namely, it no longer has the property over $(0, 1)$.

Problem 3

(a)

$$\begin{aligned}
p(\theta|x, y) &= \frac{p(y, x|\theta)p(\theta)}{p(y, x)} \\
&= \frac{p(y|x, \theta)p(x|\theta)p(\theta)}{p(y|x)p(x)} \\
&= \frac{p(y|x, \theta)p(\theta|x)}{p(y|x)}
\end{aligned}$$

$$\begin{aligned}
\theta_{\text{MAP}} &= \arg\max_\theta p(\theta|x, y) \\
&= \arg\max_\theta \frac{p(y|x, \theta)p(\theta|x)}{p(y|x)} \\
&= \arg\max_\theta p(y|x, \theta)p(\theta|x)
\end{aligned}$$

If assuming $p(\theta) = p(\theta|x)$,

$$\theta_{\text{MAP}} = \arg\max_\theta p(y|x, \theta)p(\theta)$$

(b) As $\theta \sim \mathcal{N}(0.\eta^2 I)$,

$$p(\theta) = \frac{1}{(2\pi\eta^2)^{n/2}} \exp\left(-\frac{1}{2\eta^2}\|\theta\|^2\right)$$

Hence,

$$\begin{aligned}
\theta_{\text{MAP}} &= \arg\max_\theta p(y|x, \theta)p(\theta) \\
&= \arg\min_\theta -\log\left[p(y|x, \theta) \cdot \frac{1}{(2\pi\eta^2)^{n/2}} \exp\left(-\frac{1}{2\eta^2}\|\theta\|^2\right)\right] \\
&= \arg\min_\theta \left[-\log p(y|x, \theta) - \log\frac{1}{(2\pi\eta^2)^{n/2}} + \frac{1}{2\eta^2}\|\theta\|^2\right] \\
&= \arg\min_\theta \left[-\log p(y|x, \theta) + \frac{1}{2\eta^2}\|\theta\|^2\right] \\
&= \arg\min_\theta \left[-\log p(y|x, \theta) + \lambda\|\theta\|^2\right],
\end{aligned}$$

where $\lambda = \frac{1}{2\eta^2}$.

Problem 3 continued on next page. . .

(c) Consider the loss function

$$\ell(\theta) = -\log p(y|X,\theta) + \lambda\|\theta\|^2$$

Note that $y \sim \mathcal{N}(X\theta, \sigma^2 I)$. Therefore,

$$\begin{aligned}\ell(\theta) &= -\log\left[\frac{1}{(2\pi\sigma^2)^{n/2}}\exp\left(-\frac{1}{2\sigma^2}\|y - X\theta\|^2\right)\right] + \lambda\|\theta\|^2 \\ &= -\log\frac{1}{(2\pi\sigma^2)^{n/2}} + \frac{1}{2\sigma^2}\|y - X\theta\|^2 + \lambda\|\theta\|^2\end{aligned}$$

MAP estimation is to minimize $\ell(\theta)$, which is achieved when

$$\begin{aligned}\nabla\ell(\theta) &= \frac{1}{2\sigma^2}\nabla\|y - X\theta\|^2 + \lambda\nabla\|\theta\|^2 \\ &= \frac{1}{\sigma^2}X^T(X\theta - y) + 2\lambda\theta = 0\end{aligned}$$

Thus,

$$\begin{aligned}\frac{1}{\sigma^2}X^TX\theta_{\text{MAP}} + 2\lambda\theta_{\text{MAP}} &= \frac{1}{\sigma^2}X^Ty \\ (X^TX + 2\lambda\sigma^2 I)\theta_{\text{MAP}} &= X^Ty \\ \theta_{\text{MAP}} &= (X^TX + 2\lambda\sigma^2 I)^{-1}X^Ty\end{aligned}$$

(d)

$$\begin{aligned}\theta_{\text{MAP}} &= \arg\min_\theta\left[-\log\left[\frac{1}{(2\pi\sigma^2)^{n/2}}\exp\left(-\frac{1}{2\sigma^2}\|y - X\theta\|_2^2\right)\right] - \log\left[\frac{1}{2b}\exp\left(-\frac{\|\theta\|_1}{b}\right)\right]\right] \\ &= \arg\min_\theta\left(\frac{1}{2\sigma^2}\|y - X\theta\|_2^2 + \frac{\|\theta\|_1}{b}\right) \\ &= \arg\min_\theta\left(\|y - X\theta\|_2^2 + \frac{2\sigma^2}{b}\|\theta\|_1\right) \\ &= \arg\min_\theta J(\theta),\end{aligned}$$

where $J(\theta) = \|y - X\theta\|_2^2 + \lambda\|\theta\|_1$ and $\lambda = \frac{2\sigma^2}{b}$.

Problem 4

(a) Consider $K \in \mathbb{R}^{n\times n}$ which satisfies

$$K = K_1 + K_2$$

Let $z \in \mathbb{R}^n$ be an arbitrary vector.

$$z^TKz = z^T(K_1 + K_2)z = z^TK_1z + z^TK_2z \geq 0$$

as K_1 and K_2 are both PSD. Hence K is PSD, namely, $K(x,z)$ is a kernel.

(b) Similarly to (a),

$$z^TKz = z^TK_1z - z^TK_2z,$$

which is not necessarily greater than or equal to zero. Hence, K is not necessarily PSD, and therefore, the function K is not necessarily a kernel.

(c)

$$z^TKz = az^TK_1z$$

where $a \in \mathbb{R}^+$. Hence, $K(x,z)$ is a kernel since K is PSD.

Problem 4 continued on next page...

(d) Similarly to (c), $K(x,z)$ is not a kernel since K is not PSD.

(e)

$$\begin{aligned}
w^T K w &= w^T (K_1 \odot K_2) w \\
&= \sum_i \sum_j w_i K_1(x^{(i)}, x^{(j)}) K_2(x^{(i)}, x^{(j)}) w_j \\
&= \sum_i \sum_j w_i \phi_1^T(x^{(i)}) \phi_1(x^{(j)}) \phi_2^T(^{(i)}) \phi_2(x^{(j)}) w_j \\
&= \sum_i \sum_j w_i w_j \sum_k \phi_{1,k}(x^{(i)}) \phi_{1,k}(x^{(j)}) \sum_l \phi_{2,l}(x^{(i)}) \phi_{2,l}(x^{(j)}) \\
&= \sum_k \sum_l \sum_i \sum_j w_i w_j \phi_{1,k}(x^{(i)}) \phi_{1,k}(x^{(j)}) \phi_{2,l}(x^{(i)}) \phi_{2,l}(x^{(j)}) \\
&= \sum_k \sum_l \left(\sum_i w_i \phi_{1,k}(x^{(i)}) \phi_{2,l}(x^{(i)}) \right) \left(\sum_j w_j \phi_{1,k}(x^{(j)}) \phi_{2,l}(x^{(j)}) \right) \\
&= \sum_k \sum_l \left(\sum_i w_i \phi_{1,k}(x^{(i)}) \phi_{2,l}(x^{(i)}) \right)^2 \geq 0
\end{aligned}$$

Hence, $K \succeq 0$, and therefore $K(x,z)$ is a kernel.

(f)

$$\begin{aligned}
z^T K z &= \sum_i \sum_j z_i f(x^{(i)}) f(x^{(j)}) z_j \\
&= \left(\sum_i z_i f(x^{(i)}) \right)^2 \geq 0
\end{aligned}$$

Hence, $K \succeq 0$, and therefore $K(x,z)$ is a kernel.

(g) As $K_3(x,z)$ itself is a valid kernel and we just change x and z to $\phi(x)$ and $\phi(z)$, $K(x,z)$ is a kernel.

(h) Let

$$p(x) = \sum_i a_i x^i$$

where $a_i \in [0, \infty)$. Hence,

$$K(x,z) = \sum_i a_i K_1^i(x,z)$$

From (e), it is known that the product of valid kernels is still a valid kernel, so $K_1^i(x,z)$ is a kernel. From (c), it is known that a kernel times a positive real number is still a kernel, so $a_i K_1^i(x,z)$ is a kernel. From (a), it is known that the sum of kernels is a kernel, so $K(x,z) = \sum_i a_i K_1^i(x,z)$ is a kernel.

Problem 5

(a)

i. From the update rule, we know

$$\theta^{(i)} = \sum_{j=1}^{i} \alpha \left(y^{(j)} - h_{\theta^{(j-1)}}(x^{(j)}) \right) \phi(x^{(j)})$$

And we might represent the initial value with

$$\theta^{(0)} = \sum_{j=1}^{0} \alpha \left(y^{(j)} - h_{\theta^{(j-1)}}(x^{(j)}) \right) \phi(x^{(j)}) = 0$$

ii.

$$\begin{aligned}
h_{\theta^{(i)}}(x^{(i+1)}) &= g\left((\theta^{(i)})^T \phi(x^{(i+1)}) \right) \\
&= g\left(\left(\sum_{j=1}^{i} \alpha \left(y^{(j)} - h_{\theta^{(j-1)}}(x^{(j)}) \right) \phi(x^{(j)}) \right)^T \phi(x^{(i+1)}) \right) \\
&= g\left(\sum_{j=1}^{i} \alpha \left(y^{(j)} - h_{\theta^{(j-1)}}(x^{(j)}) \right) \phi(x^{(j)})^T \phi(x^{(i+1)}) \right) \\
&= g\left(\sum_{j=1}^{i} \alpha \left(y^{(j)} - h_{\theta^{(j-1)}}(x^{(j)}) \right) K(x^{(j)}, x^{(i+1)}) \right) \\
&= \mathbf{1}\left\{ \sum_{j=1}^{i} \alpha \left(y^{(j)} - h_{\theta^{(j-1)}}(x^{(j)}) \right) K(x^{(j)}, x^{(i+1)}) \geq 0 \right\}
\end{aligned}$$

iii.

$$\begin{aligned}
&\theta^{(i+1)} := \theta^{(i)} + \alpha \left(y^{(i+1)} - h_{\theta^{(i)}}(x^{(i+1)}) \right) x^{(i+1)} \\
&= \sum_{j=1}^{i} \alpha \left(y^{(j)} - h_{\theta^{(j-1)}}(x^{(j)}) \right) \phi(x^{(j)}) + \alpha \left(y^{(i+1)} - \mathbf{1}\left\{ \sum_{j=1}^{i} \alpha \left(y^{(j)} - h_{\theta^{(j-1)}}(x^{(j)}) \right) K(x^{(j)}, x^{(i+1)}) \geq 0 \right\} \right) x^{(i+1)}
\end{aligned}$$

(b) See the code.

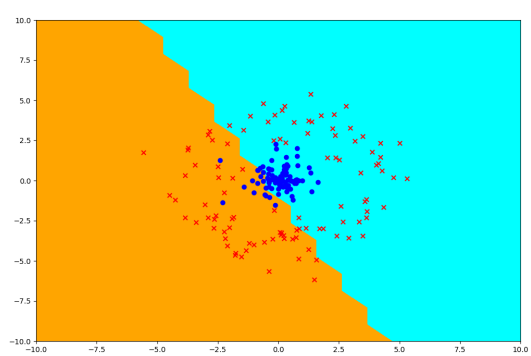

Figure 3: Kernelized perceptron with dot product kernel

(c) Obviously, the dot product kernel has much poorer performance since it is essentially linear and this dataset is non-linear-separable (see Figures 3 and 4).

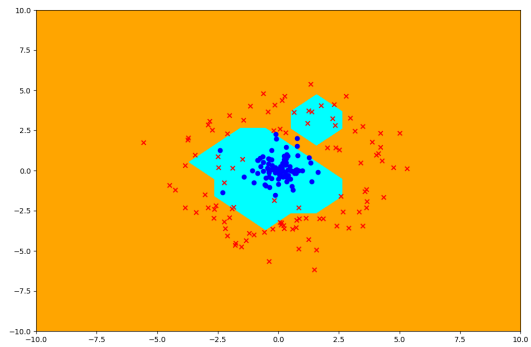

Figure 4: Kernelized perceptron with RBF kernel

Problem 6

(a) See the code.

(b) Consider an SMS as a vector $x \in \mathbb{R}^n$ that records the number of times each word in the dictionary appears. The assumption of Naive Bayes is

$$p(x_1, \cdots, x_n|y) = \prod_{i=1}^{n} p(x_i|y)$$

We assume a multinomial distribution of x. Hence, the likelihood is

$$\begin{aligned}\mathcal{L}(\phi, \phi_{j|y=0}, \phi_{j|y=1}) &= \prod_{i=1}^{m} p(x^{(i)}, y^{(i)}) \\ &= \prod_{i=1}^{m} \left(\frac{N^{(i)}!}{\prod_j x_j^{(i)}!} \prod_{j=1}^{n} \phi_{j|y^{(i)}}^{x_j^{(i)}} \right) p(y^{(i)}; \phi_y)\end{aligned}$$

To maximize the log-likelihood is to maximize

$$\ell(\phi_y, \phi_{j|y=0}, \phi_{j|y=1}) = \sum_{i=1}^{m} y^{(i)} \log(\phi_y) + (1 - y^{(i)}) \log(1 - \phi_y) + \sum_{j=1}^{n} x_j^{(i)} \log(\phi_{j|y^{(i)}})$$

Additionally, we have the constraint

$$\sum_j \phi_{j|y=1} = 1; \ \sum_j \phi_{j|y=0} = 1$$

Consider all terms with $y^{(i)} = 1$. The Lagrangian is

$$\mathcal{L}(\lambda, \phi_{j|y=1}) = -\lambda \left(1 - \sum_j \phi_{j|y=1} \right) + \sum_{i=1}^{m} \log \phi_y + y^{(i)} \sum_{j=1}^{n} x_j^{(i)} \log(\phi_{j|y=1})$$

Problem 6 continued on next page. . .

Let its gradient equal zero to maximize ℓ:

$$\begin{aligned}
\nabla \mathcal{L}(\lambda, \phi_{j|y=1}) &= 0 \\
\sum_{i=1}^{m} y^{(i)} \cdot \frac{x_j^{(i)}}{\phi_{j|y=1}} &= \lambda \\
\phi_{j|y=1} &= \frac{1}{\lambda} \sum_{i=1}^{m} y^{(i)} x_j^{(i)}
\end{aligned}$$

Note that

$$\begin{aligned}
\sum_{j} \phi_{j|y=1} &= 1 \\
\frac{1}{\lambda} \sum_{i=1}^{m} \sum_{j=1}^{n} y^{(i)} x_j^{(i)} &= 1 \\
\lambda &= \sum_{i=1}^{m} \sum_{j=1}^{n} y^{(i)} x_j^{(i)}
\end{aligned}$$

Hence,

$$\phi_{j|y=1} = \frac{\sum_{i=1}^{m} y^{(i)} x_j^{(i)}}{\sum_{i=1}^{m} \sum_{j=1}^{n} y^{(i)} x_j^{(i)}}$$

Consider the Laplace smoothing,

$$\phi_{j|y=1} = \frac{1 + \sum_{i=1}^{m} y^{(i)} x_j^{(i)}}{n + \sum_{i=1}^{m} \sum_{j=1}^{n} y^{(i)} x_j^{(i)}}$$

Similarly, $\phi_{j|y=0}$ can be written as

$$\phi_{j|y=0} = \frac{1 + \sum_{i=1}^{m} (1 - y^{(i)}) x_j^{(i)}}{n + \sum_{i=1}^{m} \sum_{j=1}^{n} (1 - y^{(i)}) x_j^{(i)}}$$

Consider maximize the likelihood w.r.t. ϕ, which is to maximize

$$\ell(\lambda, \phi_y) = \sum_{i=1}^{m} y^{(i)} \log \phi_y + (1 - y^{(i)}) \log(1 - \phi_y),$$

Let its gradient equal zero:

$$\begin{aligned}
\nabla \mathcal{L}(\lambda, \phi_y) &= 0 \\
\sum_{i=1}^{m} \frac{y^{(i)}}{\phi_y} &= \sum_{i=1}^{m} \frac{1 - y^{(i)}}{1 - \phi_y} \\
\sum_{i=1}^{m} \phi_y &= \sum_{i=1}^{m} y^{(i)} \\
\phi_y &= \frac{\sum_{i=1}^{m} y^{(i)}}{m}
\end{aligned}$$

Problem 6 continued on next page. . .

Finally, if let the length of the ith message be $N^{(i)}$ words, the probability of spam is

$$\begin{aligned}
p(y^{(i)}=1|x) &= \frac{p(x^{(i)}|y^{(i)}=1)p(y^{(i)}=1)}{p(x^{(i)}|y^{(i)}=1)p(y^{(i)}=1)+p(x^{(i)}|y^{(i)}=0)p(y^{(i)}=0)} \\
&= \frac{\left(\frac{N^{(i)}!}{\prod_{j=1}^{n} x_j^{(i)}!}\prod_{j=1}^{n}\phi_{j|y=1}^{x_j^{(i)}}\right)\phi_y}{\left(\frac{N^{(i)}!}{\prod_{j=1}^{n} x_j^{(i)}!}\prod_{j=1}^{n}\phi_{j|y=1}^{x_j^{(i)}}\right)\phi_y+\left(\frac{N^{(i)}!}{\prod_{j=1}^{n} x_j^{(i)}!}\prod_{j=1}^{n}\phi_{j|y=0}^{x_j^{(i)}}\right)(1-\phi_y)} \\
&= \frac{\left(\prod_{j=1}^{n}\phi_{j|y=1}^{x_j^{(i)}}\right)\phi_y}{\left(\prod_{j=1}^{n}\phi_{j|y=1}^{x_j^{(i)}}\right)\phi_y+\left(\prod_{j=1}^{n}\phi_{j|y=0}^{x_j^{(i)}}\right)(1-\phi_y)}
\end{aligned}$$

To avoid underflow, instead of calculating this straightaway, we use

$$\begin{aligned}
\log\frac{p(y^{(i)}=0|x^{(i)})}{p(y^{(i)}=1|x^{(i)})} &= \log\left(\frac{\left(\prod_{j=1}^{n}\phi_{j|y=0}^{x_j^{(i)}}\right)(1-\phi_y)}{\left(\prod_{j=1}^{n}\phi_{j|y=1}^{x_j^{(i)}}\right)\phi_y}\right) \\
&= \left(\sum_{j=1}^{n} x_j^{(i)}\log\phi_{j|y=0}\right)+\log(1-\phi_y)-\left(\sum_{j=1}^{n} x_j^{(i)}\log\phi_{j|y=1}\right)-\log\phi_y \\
&= \log\frac{1-\phi_y}{\phi_y}+\sum_{j=1}^{n} x_j^{(i)}\log\frac{\phi_{j|y=0}}{\phi_{j|y=1}}
\end{aligned}$$

And note that if let $z=\log\frac{p(y^{(i)}=0|x^{(i)})}{p(y^{(i)}=1|x^{(i)})}$

$$\begin{aligned}
\exp(z) &= \frac{1-p(y^{(i)}=1|x^{(i)})}{p(y^{(i)}=1|x^{(i)})} \\
\frac{1}{p(y^{(i)}=1|x^{(i)})} &= \exp(z)+1 \\
p(y^{(i)}=1|x^{(i)}) &= \frac{1}{1+\exp(z)}
\end{aligned}$$

(c) See the code.

(d) See the code.